

ELEMENTS OF FINANCIAL MANAGEMENT
EXPERIENTIAL LEARNING PROJECT - 2026

Synopsis

TITLE - Modeling Financial Contagion and Adaptive Portfolio Optimization using Graph Neural Networks		
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● **Problem Statement**

Modern financial markets form a deeply interconnected network where firms and sectors are linked through hidden financial, operational, and economic dependencies. Traditional portfolio optimization assumes asset risks are largely independent, which fails during macroeconomic shocks. In reality, distress originating in one firm or sector propagates through the network as systemic spillover, mutating across sectors and triggering widespread crashes.

Existing financial risk models can predict such spillovers or identify risky nodes, but do not translate these graph insights into autonomous, defensive portfolio management capable of reacting in real time to network-driven contagion.

Graph Neural Networks model financial markets as interconnected systems, learning how risk and shocks propagate across firms and sectors that helps achieve better accuracy.

There is a critical need for an intelligent financial management system that:

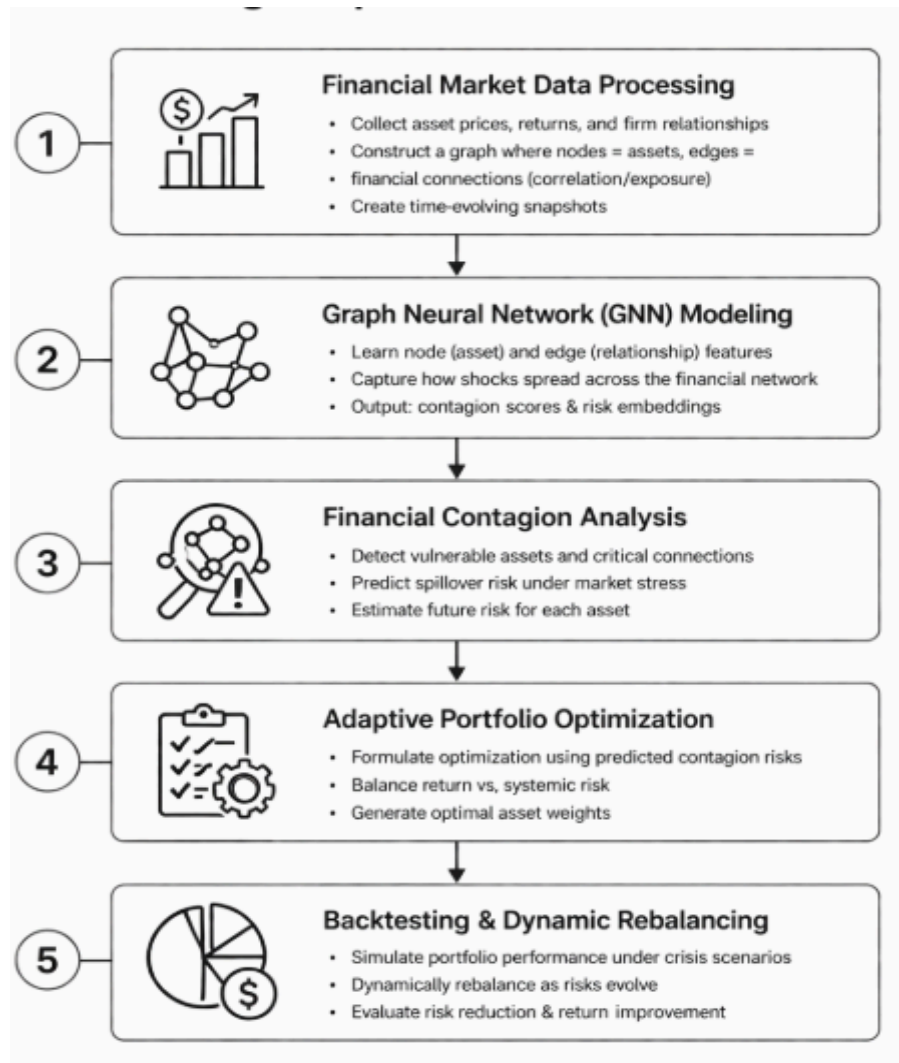
- Models systemic risk topology at both firm (micro) and sector (macro) levels
- Detects shock propagation and sector mutation early
- Performs automated portfolio rebalancing to protect assets from network-driven crashes

● **Objectives**

To design a Graph Neural Network (GNN)–driven portfolio management agent that:

1. Learns dynamic financial dependency networks between firms and sectors
2. Predicts firm-level shock propagation and inter-sector contagion
3. Simulates macroeconomic spillover effects in real time
4. Autonomously rebalances portfolio allocations toward safer sectors during systemic stress

• Methodology



• Expected Outcomes

- Early detection of systemic financial contagion before market crashes
- Identification of critical firms and sectors acting as shock amplifiers
- Real-time simulation of cross-sector mutation of financial distress
- Automated, defensive portfolio rebalancing during stress periods
- A prescriptive financial AI system that moves from risk prediction to risk prevention